

PAPER 1 / QUESTION 101

25 April 2024 / TZ2 (file label) / HL Paper 1 / original question 7 / 2224-6707

- 101.** A firework rocket launched vertically explodes into two pieces X and Y when it reaches its maximum height. The mass of X is greater than that of Y.

Three statements are made about the pieces immediately after the explosion:

- I. The kinetic energy of X is less than that of Y.
- II. The magnitude of the momentum of X is equal to that of Y.
- III. The total momentum after the explosion has increased.

Which of these statements are correct?

- A. I and II only
- B. I and III only
- C. II and III only
- D. I, II and III